

# A Gatekeeper-Constrained Dual-Stage Random Forest Surrogate for Multi-Output Ion Implantation Prediction Across Si, SiC, and GaAs

## Supplementary Materials

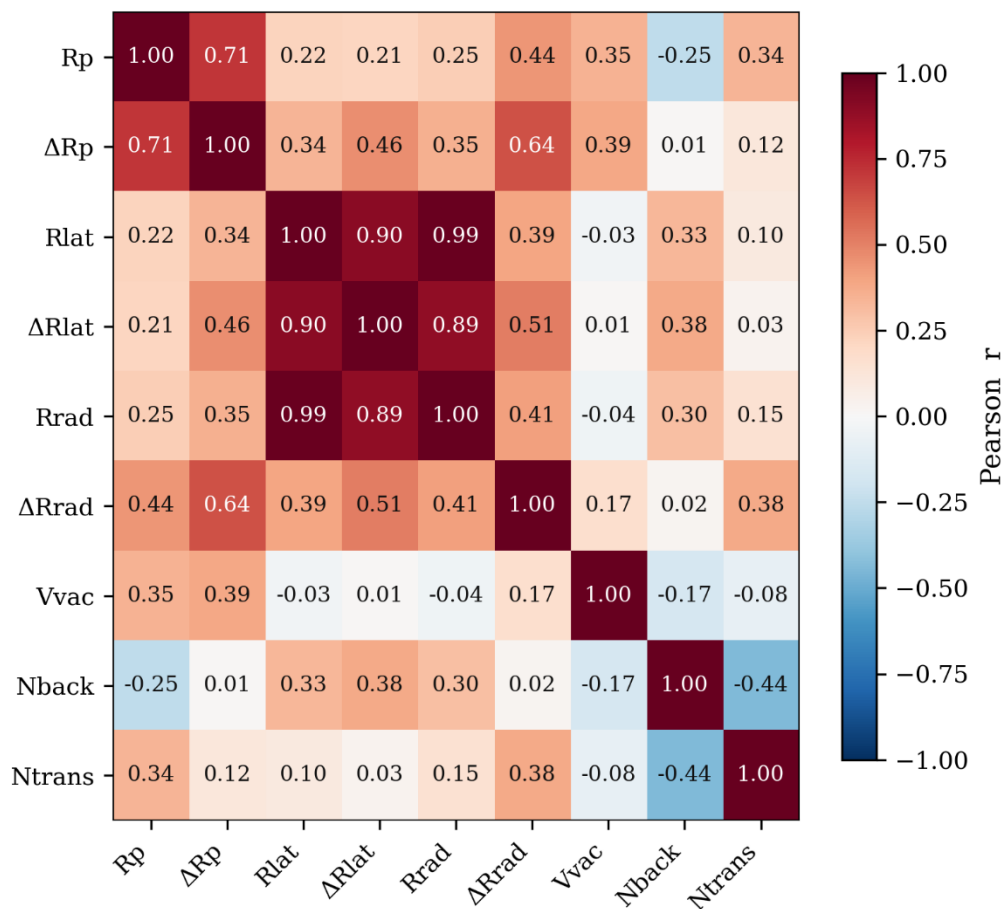


Figure S1. Pearson correlation matrix for all nine output quantities (stopping-event subset,  $n = 4,453$ ). Strong positive correlations are visible within the range group ( $R_p$ ,  $R_{lat}$ ,  $R_{rad}$ ) and straggle group ( $\Delta R_p$ ,  $\Delta R_{lat}$ ,  $\Delta R_{rad}$ ).  $N_{back}$  and  $N_{trans}$  show strong negative correlation.

Table S1:  $R^2$  as a function of ion species (ordered by atomic number Z) for three representative output quantities

| Output | Substrate | Ion | n   | R2    |
|--------|-----------|-----|-----|-------|
| Rp     | GaAs      | B   | 268 | 0.899 |
| Rp     | GaAs      | Mg  | 291 | 0.939 |
| Rp     | GaAs      | P   | 313 | 0.92  |
| Rp     | GaAs      | Ar  | 319 | 0.924 |
| Rp     | GaAs      | As  | 332 | 0.985 |
| Rp     | Si        | B   | 247 | 0.885 |
| Rp     | Si        | Mg  | 289 | 0.943 |
| Rp     | Si        | P   | 301 | 0.867 |
| Rp     | Si        | Ar  | 305 | 0.952 |
| Rp     | Si        | As  | 305 | 0.983 |
| Rp     | SiC       | B   | 260 | 0.904 |
| Rp     | SiC       | Mg  | 297 | 0.934 |
| Rp     | SiC       | P   | 314 | 0.94  |
| Rp     | SiC       | Ar  | 308 | 0.972 |
| Rp     | SiC       | As  | 304 | 0.981 |
| DRp    | GaAs      | B   | 268 | 0.837 |
| DRp    | GaAs      | Mg  | 291 | 0.84  |
| DRp    | GaAs      | P   | 313 | 0.859 |
| DRp    | GaAs      | Ar  | 319 | 0.894 |
| DRp    | GaAs      | As  | 332 | 0.958 |
| DRp    | Si        | B   | 247 | 0.63  |
| DRp    | Si        | Mg  | 289 | 0.712 |
| DRp    | Si        | P   | 301 | 0.768 |
| DRp    | Si        | Ar  | 305 | 0.853 |
| DRp    | Si        | As  | 305 | 0.819 |
| DRp    | SiC       | B   | 260 | 0.478 |
| DRp    | SiC       | Mg  | 297 | 0.52  |
| DRp    | SiC       | P   | 314 | 0.744 |
| DRp    | SiC       | Ar  | 308 | 0.694 |
| DRp    | SiC       | As  | 304 | 0.763 |
| Vvac   | GaAs      | B   | 268 | 0.91  |
| Vvac   | GaAs      | Mg  | 291 | 0.949 |
| Vvac   | GaAs      | P   | 313 | 0.963 |
| Vvac   | GaAs      | Ar  | 319 | 0.944 |
| Vvac   | GaAs      | As  | 332 | 0.94  |
| Vvac   | Si        | B   | 247 | 0.914 |
| Vvac   | Si        | Mg  | 289 | 0.951 |
| Vvac   | Si        | P   | 301 | 0.949 |
| Vvac   | Si        | Ar  | 305 | 0.942 |

|      |     |    |     |       |
|------|-----|----|-----|-------|
| Vvac | Si  | As | 305 | 0.955 |
| Vvac | SiC | B  | 260 | 0.943 |
| Vvac | SiC | Mg | 297 | 0.949 |
| Vvac | SiC | P  | 314 | 0.942 |
| Vvac | SiC | Ar | 308 | 0.931 |
| Vvac | SiC | As | 304 | 0.939 |

Table S2. Experimental fidelity: RF surrogate vs. SRIM-2013 vs. published experimental data (28 retained conditions from four independent sources).

| Sub. | Ion | $\theta$ (°) | E (keV) | Exp. Rp (Å) | SRIM Rp (Å) | RF Rp (Å) | SRIM Error% | RF Error% | Source             |
|------|-----|--------------|---------|-------------|-------------|-----------|-------------|-----------|--------------------|
| SiC  | B   | 7°           | 10      | 293         | 270         | 269       | 7.8%        | 8.2%      | Janson 2003 [42]   |
| SiC  | B   | 7°           | 100     | 1686        | 1998        | 1995      | 18.5%       | 18.3%     | Janson 2003 [42]   |
| SiC  | P   | 7°           | 50      | 411         | 464         | 465       | 8.3%        | 13.1%     | Janson 2003 [42]   |
| SiC  | P   | 7°           | 100     | 806         | 890         | 899       | 10.4%       | 11.5%     | Janson 2003 [42]   |
| SiC  | P   | 7°           | 300     | 2339        | 2498        | 2488      | 6.8%        | 6.4%      | Janson 2003 [42]   |
| SiC  | P   | 7°           | 500     | 3840        | 3974        | 3997      | 3.5%        | 4.1%      | Janson 2003 [42]   |
| SiC  | P   | 7°           | 800     | 6058        | 5886        | 5864      | 2.8%        | 3.2%      | Janson 2003 [42]   |
| SiC  | P   | 7°           | 1000    | 7521        | 6998        | 6893      | 7.0%        | 8.3%      | Janson 2003 [42]   |
| GaAs | As  | 7°           | 540     | 2000        | 2349        | 2193      | 17.4%       | 9.7%      | Landgren 1988 [45] |
| SiC  | As  | 12.5°        | 40      | 270         | 222         | 270       | 17.8%       | 0.0%      | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 60      | 373         | 307         | 332       | 17.7%       | 11.0%     | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 80      | 471         | 388         | 418       | 17.6%       | 11.3%     | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 100     | 566         | 468         | 481       | 17.3%       | 15.0%     | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 150     | 787         | 679         | 753       | 13.7%       | 4.3%      | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 200     | 991         | 885         | 966       | 10.7%       | 2.5%      | Ahmed 1995 [43]    |
| SiC  | As  | 12.5°        | 300     | 1380        | 1291        | 1309      | 6.4%        | 5.1%      | Ahmed 1995 [43]    |
| SiC  | B   | 12.5°        | 40      | 1025        | 917         | 1119      | 10.5%       | 9.2%      | Ahmed 1995 [43]    |
| SiC  | B   | 12.5°        | 60      | 1429        | 1283        | 1350      | 10.2%       | 5.5%      | Ahmed 1995 [43]    |
| SiC  | B   | 12.5°        | 80      | 1814        | 1637        | 1594      | 9.8%        | 12.1%     | Ahmed 1995 [43]    |
| SiC  | B   | 12.5°        | 150     | 3050        | 2789        | 2888      | 8.6%        | 5.3%      | Ahmed 1995 [43]    |
| SiC  | B   | 12.5°        | 250     | 4640        | 4234        | 4047      | 8.8%        | 10.0%     | Ahmed 1995 [43]    |
| Si   | P   | 0°           | 100     | 1200        | 1365        | 1368      | 13.8%       | 14.0%     | Crowder 1971 [44]  |
| Si   | P   | 0°           | 140     | 1800        | 1864        | 1999      | 3.6%        | 11.1%     | Crowder 1971 [44]  |
| Si   | P   | 0°           | 200     | 2800        | 2615        | 2596      | 6.6%        | 7.3%      | Crowder 1971 [44]  |
| Si   | P   | 0°           | 280     | 3800        | 3610        | 3631      | 5.0%        | 4.4%      | Crowder 1971 [44]  |
| Si   | As  | 0°           | 80      | 600         | 594         | 592       | 1.0%        | 1.3%      | Crowder 1971 [44]  |
| Si   | As  | 0°           | 140     | 950         | 972         | 1069      | 2.3%        | 12.5%     | Crowder 1971 [44]  |
| Si   | As  | 0°           | 280     | 1800        | 1838        | 1930      | 2.1%        | 7.2%      | Crowder 1971 [44]  |

Table S3. Computational performance benchmark: SRIM-2013 vs RF surrogate  
(5 ions  $\times$  3 substrates, 10,000 ions per run)

| Ion | Substrate | Thickness (Å) | Density (g cm <sup>-3</sup> ) | Energy (keV) | Angle (°) | N ions | SRIM time, i5 (s) | SRIM time, i9 (s) | RF inference (ms) | Speedup vs i5 | Speedup vs i9 |
|-----|-----------|---------------|-------------------------------|--------------|-----------|--------|-------------------|-------------------|-------------------|---------------|---------------|
| Ar  | Si        | 10000         | 2.329                         | 100          | 7         | 10000  | 299.74            | 156.6             | 465.3855          | 644.0682      | 336.4952      |
| As  | Si        | 10000         | 2.329                         | 100          | 7         | 10000  | 186.7             | 67.7              | 453.4028          | 411.7752      | 149.3154      |
| B   | Si        | 10000         | 2.329                         | 100          | 7         | 10000  | 267.77            | 134.57            | 459.0701          | 583.2878      | 293.1361      |
| P   | Si        | 10000         | 2.329                         | 100          | 7         | 10000  | 294.81            | 155               | 479.4629          | 614.8755      | 323.2784      |
| Mg  | Si        | 10000         | 2.329                         | 100          | 7         | 10000  | 448.3             | 284.3             | 487.2821          | 920.0009      | 583.4402      |
| Ar  | SiC       | 10000         | 3.21                          | 100          | 7         | 10000  | 699.81            | 439.81            | 462.8002          | 1512.121      | 950.3238      |
| As  | SiC       | 10000         | 3.21                          | 100          | 7         | 10000  | 294.41            | 154.41            | 477.8894          | 616.063       | 323.1082      |
| B   | SiC       | 10000         | 3.21                          | 100          | 7         | 10000  | 203.72            | 104.72            | 474.9275          | 428.9497      | 220.4968      |
| P   | SiC       | 10000         | 3.21                          | 100          | 7         | 10000  | 331.83            | 131.83            | 507.6985          | 653.5966      | 259.662       |
| Mg  | SiC       | 10000         | 3.21                          | 100          | 7         | 10000  | 439.9             | 239.9             | 504.2242          | 872.4295      | 475.7805      |
| Ar  | GaAs      | 10000         | 5.316                         | 100          | 7         | 10000  | 265.79            | 125.79            | 457.1229          | 581.441       | 275.1776      |
| As  | GaAs      | 10000         | 5.316                         | 100          | 7         | 10000  | 306.32            | 156.32            | 449.9726          | 680.7526      | 347.399       |
| B   | GaAs      | 10000         | 5.316                         | 100          | 7         | 10000  | 419.9             | 219.9             | 458.069           | 916.674       | 480.0586      |
| P   | GaAs      | 10000         | 5.316                         | 100          | 7         | 10000  | 241.26            | 121.26            | 458.5557          | 526.1302      | 264.439       |
| Mg  | GaAs      | 10000         | 5.316                         | 100          | 7         | 10000  | 337.34            | 137.34            | 453.5955          | 743.7022      | 302.7808      |